

7. Integration Techniques

7.2. Trigonometric Integrals

7.2 Learning Objectives

- I combine methods from previous sections to find general strategies to solve trigonometric integrals.

Example 7.2.1

$$\int \cos^3(x) \, dx$$

Note

With our current techniques, we can try to recognize what function has a derivative of this, try a u -substitution, or integration by parts (or reduction formula). We don't clearly know this antiderivative, and we can't use u -sub because there isn't a spot for du . Lastly, we could try integrating by parts and/or reduction formula, but that will be incredibly tedious. Let's try the alternative in the solution below.

Whenever we see a power of $\cos(x)$ (or $\sin(x)$) of 2 or more we can try to use the Pythagorean Identity:

Theorem 7.2.1 Pythagorean Identity

For any angle x we have

$$\sin^2(x) + \cos^2(x) = 1$$

This can be rearranged by dividing by $\sin^2(x)$ or $\cos^2(x)$:

$$1 + \cot^2(x) = \csc^2(x)$$

and

$$\tan^2(x) + 1 = \sec^2(x).$$

Solution

Recall the Pythagorean Identity $\sin^2(x) + \cos^2(x) = 1$, rearranging this we can write $\cos^2(x) = 1 - \sin^2(x)$. Then using this in the integral we have

$$\begin{aligned} \int \cos^3(x) \, dx &= \int \cos^2(x) \cos(x) \, dx \\ &= \int (1 - \sin^2(x)) \cos(x) \, dx \end{aligned}$$

This might look more complicated at first, but now if we tried a u -sub it will work! Let $u = \sin(x)$, then $du = \cos(x) \, dx$, so the integral becomes

$$\begin{aligned}
 \int (1 - \sin^2(x)) \cos(x) \, dx &= \int 1 - u^2 \, du \\
 &= u - \frac{u^3}{3} \\
 &= \sin(x) - \frac{\sin^3(x)}{3}
 \end{aligned}$$

Let's check our answer by taking its derivative:

$$\begin{aligned}
 \frac{d}{dx} \sin(x) - \sin^3 \frac{x}{3} &= \cos(x) - \sin^2(x) \cos(x) \\
 &= \cos(x)(1 - \sin^2(x)) \\
 &= \cos(x) \cos^2(x) \\
 &= \cos^3(x)
 \end{aligned}$$

Note we used the same Pythagorean Identity here to get back to the same thing we started with.

Warning

In this section and on its WebAssign problems you may have to use other trigonometric identities. It is good to remember these exist every now and then, and practice using them, but having you memorize too many of these is not worth your time and energy.

You are expected to know the Pythagorean identity and alternative versions of it, but any other identity you need you may look up. Examples include the half angle and double angle formulas, and product to sum formulas, etc.

Notice in the above problem that in our u -sub, it was nice to have everything in terms of 1 trig function ($\sin(x)$ in the above example), except a single multiple of the other trig function times dx ($\cos(x)$ in the above example). Then the first variable can be our u . We will repeat this technique below, getting rid of most of one trig function except something to work for du . This takes a lot of practice and trial and error to get good at, make sure to try things and be okay failing and trying again.

Example 7.2.2

Find

$$\int_0^{\frac{\pi}{2}} \sin^5(x) \cos^2(x) \, dx.$$

Because we need 1 trig function left over for du , and the power of $\sin(x)$ is odd, we can leave 1 $\sin(x)$ alone, and replace the other 4 powers with $\cos(x)$ through the Pythagorean identity.

 Solution

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^5(x) \cos^2(x) \, dx &= \int_0^{\frac{\pi}{2}} \sin^4(x) \sin(x) \cos^2(x) \, dx \\ &= \int_0^{\frac{\pi}{2}} (1 - \cos^2(x))^2 \sin(x) \cos^2(x) \, dx\end{aligned}$$

And now a u -sub works! Let $u = \cos(x)$, then $du = -\sin(x) \, dx$, and $u(0) = 1$ and $u(\frac{\pi}{2}) = 0$.

So we have

$$\begin{aligned}\int_0^{\frac{\pi}{2}} (1 - \cos^2(x))^2 \sin(x) \cos^2(x) \, dx &= \int_1^0 (1 - u^2)^2 u^2 (-1) \, du \\ &= \int_0^1 (1 - u^2)^2 u^2 \, du \\ &= \int_0^1 u^2 - 2u^4 + u^6 \, du \\ &= \left[\frac{u^3}{3} - \frac{2}{5}u^5 + \frac{u^7}{7} \right]_0^1 \\ &= \frac{1}{3} - \frac{2}{5} + \frac{1}{7}\end{aligned}$$

Example 7.2.3

Find

$$\int \sin^2(x) \cos^2(x) \, dx.$$

Here, we have an even number on both exponents, so using the Pythagorean identity wouldn't leave anything for du at the end of the rearranging. We need a new strategy.

Theorem 7.2.2 Double Angle Formula

For any angle x we have

$$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

and

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

 Solution

When we use double angle formula, we might have to simplify a lot and try the formula again.

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \int \frac{1}{2}(1 - \cos(2x)) \frac{1}{2}(1 + \cos(2x)) \, dx \\ &= \frac{1}{4} \int 1 - \cos^2(2x) \, dx\end{aligned}$$

Then we have to use double angle formula again.

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \frac{1}{4} \int 1 - \cos^2(2x) \, dx \\ &= \frac{1}{4} \int 1 - \frac{1}{2}(1 + \cos(4x)) \, dx \\ &= \frac{1}{4} \int 1 - \frac{1}{2} - \cos(4x) \, dx\end{aligned}$$

Then let $u = 4x$, $du = 4 \, dx$ so

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \frac{1}{4} \int 1 - \frac{1}{2} - \frac{\cos(4x)}{2} \, dx \\ &= \frac{1}{4} \int \frac{1}{2} \, dx - \frac{1}{32} \int \cos(u) \, du \\ &= \frac{1}{8}x - \frac{1}{32} \sin(4x) + C.\end{aligned}$$

Tip

Strategy for evaluating $\int \sin^m(x) \cos^n(x) \, dx$

- **Case 1: At least one of m or n are odd:**
 - Choose the trig function with the odd power to be u . If both are odd, just pick one.
 - Save one factor of this trig function for du later.
 - Rewrite the other trig function (even power) with the Pythagorean Identity.
 - Do u -sub, expand everything, and evaluate.
- **Case 2: Both m and n are even**
 - Reduce all even powers with the double angle identities.
 - You may need to expand and apply the previous step multiple times until all even powers are eliminated.
 - The identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ may sometimes be useful.

Example 7.2.4

Find

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \tan^5(x) \sec^7(x) \, dx$$

Building off what we know from $\sin(x)$ and $\cos(x)$, we will probably want to pick u so that du is still around. Recall the derivatives of the 4 non sinusoidal trig functions

 Note

You should know the following derivatives from the prerequisites to this class, if you are unfamiliar with these, please review and/or discuss in office hours.

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

 Solution

Now, when we pick u we should plan to “save” the derivative of u and replace everything else so it is in terms of u . We can either keep a $\sec^2(x)$ around, or $\sec(x) \tan(x)$. Depending on if u is going to be $\tan(x)$ or $\sec(x)$. Either way should work, we will pick $u = \sec(x)$ here, try $u = \tan(x)$ on your own as practice.

Let $u = \sec(x)$, then $du = \sec(x) \tan(x) dx$. This gets us

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \tan^4(x) \underbrace{\sec^6(x)}_{u^6} \underbrace{\tan(x) \sec(x) dx}_{du}$$

This leaves us with $\tan^4(x)$ that we have to rewrite with Pythagorean identity. Then, using $\tan^2(x) = \sec^2(x) - 1$ we have

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\sec^2(x) - 1)^2 \sec^6(x) \tan(x) \sec(x) dx &= \int_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} (u^2 - 1)^2 u^6 du \\ &= \int_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} u^{10} - 2u^8 + 1u^6 du \\ &= \left[\frac{u^{11}}{11} - \frac{2}{9}u^9 + \frac{u^7}{7} \right]_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} \end{aligned}$$

Simplifying this expression is left as an exercise to the reader.

Example 7.2.5

Find

$$\int \cot^2(x) \csc^4(x) dx$$

Solution

Let $u = \cot(x)$, then $du = -\csc^2(x) dx$. Then,

$$\int \cot^2(x) \csc^4(x) dx = \int (-1) \underbrace{\cot^2(x)}_{u^2} \underbrace{\csc^2(x)}_{\text{rewrite}} \underbrace{(-1) \csc^2(x) dx}_{du}$$

Then using the Pythagorean Identity $\csc^2(x) = 1 + \cot^2(x)$ we have

$$\begin{aligned} \int \cot^2(x) \csc^4(x) dx &= - \int \cot^2(x) (1 + \cot^2(x))^2 (-\csc^2(x)) dx \\ &= - \int u^2 (1 + u^2)^2 du \\ &= - \int u^2 + 2u^4 + u^6 du \\ &= - \left[\frac{u^3}{3} + \frac{2}{5} u^5 + \frac{u^7}{7} \right] + C \\ &= - \left[\frac{\cot^3(x)}{3} + \frac{2}{5} \cot^5(x) + \frac{\cot^7(x)}{7} \right] + C \end{aligned}$$

Tip

Strategy for evaluating $\int \tan^m(x) \sec^n(x) dx$

- **Case 1: n even:** (i.e. the power of secant is even)
 - Save one factor of $\sec^2(x)$ for du later.
 - Rewrite the rest of the powers of $\sec^2(x)$ using $\sec^2(x) = 1 + \tan^2(x)$ (even power) with the Pythagorean Identity.
 - Do u -sub, expand everything, and evaluate.
- **Case 2: m is odd:** (i.e. the power of tangent is odd)
 - Save one factor of $\sec(x) \tan(x)$ for du later.
 - Convert the remaining even powers of tangent to $\tan^2(x) = \sec^2(x) - 1$
 - Do u -sub, expand everything, and evaluate.
- You may also use the following 2 antiderivatives

$$\int \tan(u) du = \ln|\sec(u)| + C$$

$$\int \sec(u) du = \ln|\sec(u) + \tan(u)| + C$$

Exercise 7.2.1

Derive the above antiderivatives. The second needs a clever multiplication by 1 trick. Come to office hours if you want to go through this together.

7.2 Section Summary:

- We used trigonometric identities and other tools to rearrange trigonometric integrals to be able to solve them.